Exercises: Linear Systems and Solution Theory

Mathematical Foundations for Machine Learning

Subspaces, Span, and Kernel

Exercise 1.1 [Easy]

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{pmatrix}$.

a) Compute $\ker(A)$.
b) Give a basis for $\mathrm{im}(A)$.
c) Verify the rank-nullity theorem for A.

Exercise 1.2 [Medium]

Let $U, W \subseteq \mathbb{R}^n$ be subspaces.

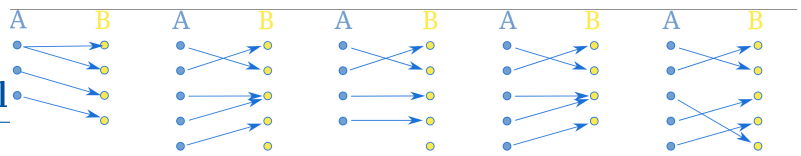
a) Prove that $U \cap W$ is a subspace of $\mathbb{R}^n$.
b) Is $U \cup W$ necessarily a subspace? Either prove it or give a counterexample.
c) In a linear regression setting, suppose two groups of features $x_A \in \mathbb{R}^{d_1}$ and $x_B \in \mathbb{R}^{d_2}$ produce predictions via $y_A = Ax_A$ and $y_B = Bx_B$ respectively, where $A \in \mathbb{R}^{m \times d_1}$ and $B \in \mathbb{R}^{m \times d_2}$. Using part (a), explain why the set of predictions achievable by *both* feature groups is a subspace, and express it in terms of A and B. img and Ker of a subspace under linear map are both subspaces

Exercise 1.3 [Hard]

Let $A \in \mathbb{R}^{m \times n}$. Prove that $\ker(A^T A) = \ker(A)$.

Hint: One inclusion is straightforward. For the other, consider $\|Ax\|^2$.

(**ML context:** This identity explains why the normal equations $A^T A\theta = A^T y$ have the same solution set structure as the original system $A\theta = y$ in least-squares regression.)



if dim(V) = dim(W) then injective = surjective = bij.you need bijective for inverse

Rank, Image, and the Rank-Nul

Exercise 2.1 [Easy]

A linear map $T : \mathbb{R}^7 \to \mathbb{R}^4$ has $\mathrm{rank}(T) = 3$.

a) What is $\dim(\ker(T))$?
b) Is T surjective? Is T injective? Justify each answer.
c) How many free parameters does the general solution of $Tx = b$ have (when a solution exists)?

T is injective $\iff \ker(T) = \{0\} \iff \dim(\ker(T)) = 0$

else all balls of kernel point to the same value 0

$T(x_0 + v) = T(x_0) + T(v) = b + 0 = b$

we have ker(t) free params -> a single solution in the codomain could come from a domain of span ker(t)

Exercise 2.2 [Medium]

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$.

a) Show that $\mathrm{im}(AB) \subseteq \mathrm{im}(A)$.
b) Conclude that $\mathrm{rank}(AB) \leq \min\{\mathrm{rank}(A), \mathrm{rank}(B)\}$.

Hint for the second inequality: use part (a) applied to $(AB)^T = B^T A^T$. $\mathrm{rank}(M) = \mathrm{rank}(M^T)$

Exercise 2.3 [Hard]

Let $P \in \mathbb{R}^{n \times n}$ satisfy $P^2 = P$ (such a matrix is called a *projection*).

a) Show that $\mathrm{im}(P) = \ker(I - P)$ and $\ker(P) = \mathrm{im}(I - P)$.

b) Show that $\mathbb{R}^n = \mathrm{im}(P) \oplus \ker(P)$, i.e., every $x \in \mathbb{R}^n$ can be written uniquely as $x = u + v$ with $u \in \mathrm{im}(P)$ and $v \in \ker(P)$.

c) What are the possible eigenvalues of P? Use part (a) to justify.

d) Conclude that $\mathrm{rank}(P) = \mathrm{tr}(P)$.

(ML context: The hat matrix $H = X(X^TX)^{-1}X^T$ in linear regression is a projection. Its trace $\mathrm{tr}(H) = d$ equals the number of features, which measures model complexity.)

Solution Theory of Linear Systems

Exercise 3.1 [Easy]

Consider the system $Ax = b$ with

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}.$$

a) Form the augmented matrix $(A \mid b)$ and reduce it to row echelon form.

b) Does the system have a solution? If so, is it unique? Justify using the solution theory (i.e., check whether $b \in \mathrm{im}(A)$ and whether $\ker(A) = \{0\}$).

c) Write the full solution set in the form $w + \ker(A)$.

particular solution + ker(A) = general solution
bec. adding T(v + ker(A)) = T(v) + T(ker(A)) = T(v) + 0

Exercise 3.2 [Medium]

Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$. Let $B \in \mathbb{R}^{m\times m}$ be an invertible matrix.

a) Prove that $Ax = b$ and $BAx = Bb$ have the same solution set.

b) Explain why this result justifies Gaussian elimination: every elementary row operation is a multiplication from the left by an invertible matrix, so the solution set is preserved.

c) Now let $C \in \mathbb{R}^{n\times n}$ be invertible. Does $Ax = b$ and $AC^{-1}y = b$ have the same solution set? If not, what is the precise relationship between the solutions?

if x solves (1), then y=Cx solves (2), but they are different vectors (there is a bijection between solutions but diff)

Exercise 3.3 [Medium]

Let $A \in \mathbb{R}^{m\times n}$. Consider the linear system $Ax = b$.

a) Show that the solution set of $Ax = b$ (when nonempty) is an affine subspace, i.e., it has the form $w + V$ where V is a subspace of $\mathbb{R}^n$. To proof S1 inside affine subspace, search ker(A) with def Aw=b

b) Conversely, show that any affine subspace of $\mathbb{R}^n$ can be written as the solution set of some linear system.

Hint for (b): If V is a k-dimensional subspace, let $v_1, \ldots, v_k$ be a basis and construct a suitable matrix.

$$a_i^T v = a_i^T(\lambda_1 v_1 + \cdots + \lambda_k v_k) = \lambda_1 \underbrace{a_i^T v_1}_{=0} + \cdots + \lambda_k \underbrace{a_i^T v_k}_{=0} = 0$$

Exercise 3.4 [Hard]

Consider a neural network layer performing the affine transformation $y = Wx + b$ where $W \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$. Suppose $\mathrm{rank}(W) = r < \min\{m, n\}$.

a) Show that the set of all inputs $x \in \mathbb{R}^n$ that produce a given output $y_0 \in \mathrm{im}(W) + b$ is an affine subspace of dimension $n - r$.

b) An adversarial perturbation δ is *undetectable by the layer* if $W(x+\delta)+b = Wx+b$. Show that the set of undetectable perturbations equals $\ker(W)$ and has dimension $n - r$.

c) A practitioner argues: "To reduce vulnerability to adversarial perturbations, we should use layers with $\mathrm{rank}(W) = n$." Under what condition on m and n is this possible? What trade-off does this impose on the network architecture?

Gaussian Elimination and LU Decomposition

Exercise 4.1 [Easy]

Bring the following matrix to row echelon form using Gaussian elimination. For each step, write down the corresponding elementary row transformation matrix.

$$A = \begin{pmatrix} 2 & 1 & -1 \\ -6 & -2 & 4 \\ 4 & 3 & -1 \end{pmatrix}$$

In LU decomposition don't multiply rows by scalars -> only add multiples of other rows

a) Perform the elimination and write A in row echelon form U.
b) Using the elementary matrices, write $A = LU$ where L is lower triangular.
c) What is $\det(A)$?

Exercise 4.2 [Medium]

gaussian elim (n): each row (n rows) * eliminate n rows below (n rows) * cost of elim (n) * n vectors

$$\underbrace{O(n^3)}_{\text{LU once}} + \underbrace{3 \times (O(n^2) + O(n^2))}_{\text{3 systems, each forward + back}} = O(n^3) + n \times O(n^2) = O(n^3)$$

Let $A \in \mathbb{R}^{n \times n}$ be invertible with LU decomposition $A = LU$.

a) Describe the algorithm for computing A^{-1} using the LU decomposition. How many forward and backward substitutions are needed?

b) What is the total arithmetic complexity of computing A^{-1} via LU, including the cost of the decomposition itself? Express your answer in terms of n.

c) Suppose you need to solve $Ax = b_i$ for k different right-hand sides $b_1, \ldots, b_k$. Compare the cost of:

(i) Computing A^{-1} and then multiplying $x_i = A^{-1}b_i$ for each i.

(ii) Reusing the LU decomposition to solve each system.

For what values of k (relative to n) does strategy (i) become more efficient?

Exercise 4.3 [Medium]

Consider a matrix $A \in \mathbb{R}^{4 \times 5}$ whose row echelon form is:

$$U = \begin{pmatrix} 3 & 1 & 0 & 2 & -1 \\ 0 & 0 & 5 & 1 & 3 \\ 0 & 0 & 0 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

a) What is $\text{rank}(A)$?
b) Identify the pivot and free variables. What is $\dim(\ker(A))$?
c) Parameterize $\ker(A)$ by finding a basis (solve $Ux = 0$ using backward substitution with the free variables as parameters).
d) For which vectors $b \in \mathbb{R}^4$ does $Ax = b$ have a solution? Express the condition in terms of the components of b after applying the same row operations that reduce A to U.

Block Structure and Inverse via LU

Exercise 5.1 [Medium]

Compute the inverse of

$$A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$$

using the LU decomposition. Show all steps: find L and U, then solve $Lz_i = e_i$ and $Ua_i^{-1} = z_i$ for $i = 1, 2$.

first compute y and then x

Exercise 5.2 [Medium]

Consider the block-structured system:

$$\begin{pmatrix} D & B \\ C & A_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

where $D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $C = \begin{pmatrix} 1 & 1 \end{pmatrix}$, $A_{22} = (2)$, $b_1 = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$, and $b_2 = (4)$.

a) Compute the Schur complement $S = A_{22} - CD^{-1}B$.

b) Solve for x_2 using $Sx_2 = b_2 - CD^{-1}b_1$.

c) Back-substitute to find x_1.

d) Explain why exploiting the block structure is advantageous when D is a large diagonal matrix.

Mixed Challenging Exercises

Exercise 6.1 [Hard]

Let $A \in \mathbb{R}^{m \times n}$ with $m < n$ and $\operatorname{rank}(A) = m$ (full row rank).

a) Show that $Ax = b$ has a solution for every $b \in \mathbb{R}^m$.

b) Show that the solution is never unique.

c) Among all solutions, the *minimum-norm solution* is $x^* = A^T(AA^T)^{-1}b$. Show that $Ax^* = b$ and that $x^* \in \operatorname{im}(A^T)$.

d) Show that any solution $x = x^* + v$ with $v \in \ker(A)$, $v \neq 0$, satisfies $\|x\|^2 > \|x^*\|^2$.

Hint for (d): use the fact that $\operatorname{im}(A^T)$ *and* $\ker(A)$ *are orthogonal complements.*

Exercise 6.2 [Hard]

In ridge regression, we solve $(X^TX + \lambda I)\theta = X^Ty$ for a regularization parameter $\lambda > 0$ and data matrix $X \in \mathbb{R}^{N \times d}$.

a) Show that $(X^TX + \lambda I)$ is always invertible for $\lambda > 0$, regardless of the rank of X.

b) What is $\ker(X^TX + \lambda I)$? What does this imply about the uniqueness of the ridge regression solution?

c) Show that the ridge solution can equivalently be obtained by solving the augmented system:

$$\begin{pmatrix} X \\ \sqrt{\lambda}\, I \end{pmatrix} \theta = \begin{pmatrix} y \\ 0 \end{pmatrix}$$

in the least-squares sense (i.e., minimizing $\|\tilde{X}\theta - \tilde{y}\|^2$ for the augmented matrix and vector).

d) What is the rank of the augmented matrix $\tilde{X} = \begin{pmatrix} X \\ \sqrt{\lambda} I \end{pmatrix} \in \mathbb{R}^{(N+d) \times d}$ when $\lambda > 0$? Use the solution theory to explain why the least-squares solution is unique.

Exercise 6.3 [Hard]

Let $A \in \mathbb{R}^{n \times n}$ and suppose Gaussian elimination *without* row swaps produces the LU decomposition $A = LU$.

a) Show that all leading principal minors of A are nonzero. (The k-th leading principal minor is $\det(A_{1:k,1:k})$.)

Hint: Use the block structure of L and U restricted to the first k rows and columns.

b) Conversely, suppose all leading principal minors of A are nonzero. Explain (without full proof) why this guarantees that Gaussian elimination can proceed without row swaps.

c) Give a 2×2 invertible matrix that requires a row swap (i.e., does not admit an LU decomposition without permutation). Verify that its first leading principal minor is zero.